# ImageEval 2026: Culturally Grounded Arabic Multimodal Evaluation

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https://imageeval2026.github.io/

## Abstract

We present an overview of the **ImageEval 2026** shared task on culturally grounded Arabic multimodal evaluation. It includes two tasks: (*i*) **AynVQA**, covering spoken visual question answering and image-grounded hallucination detection in English and Modern Standard Arabic (MSA), and (*ii*) **CRAI-Bench**, evaluating the cultural accuracy of text-to-image generation. A total of 14 teams participated in the test phase, with 12 teams submitting system description papers. Participating systems used a range of approaches, including zero-shot prompting, fine-tuning of vision-language models, speech-recognition pipelines, ensembling, and score calibration. We describe the task setup, datasets, evaluation procedure, and participating systems, and summarize the main results across the different tracks. All datasets and evaluation scripts from the shared task are released to the research community. The shared task highlights challenges in culturally grounded multimodal evaluation, particularly for Arabic speech and image-text reasoning.

## 1 Introduction

Recent benchmarking efforts have evaluated the capabilities of LLMs, speech-language models, and vision-language models (VLMs) across tasks such as spoken question answering (SQA) (Lee et al., 2018) and visual question answering (VQA) (Hudson and Manning, 2019; Goyal et al., 2017). These benchmarks have driven substantial progress in multimodal reasoning and compositional understanding. However, most focus on general scene understanding and provide limited evidence of models' ability to reason about culturally specific concepts or remain robust across languages and regional varieties. Cross-lingual benchmarks such as xGQA extend VQA beyond English, yet largely preserve the same underlying visual domain (Pfeiffer et al., 2022).

To address this limitation, culture-centered benchmarks such as CulturalVQA (Nayak et al., 2024), CVQA (Romero et al., 2024), SEA-VQA (Urailertprasert et al., 2024), and OASIS (Alam et al., 2025b) broaden evaluation to culturally situated concepts, including artifacts, food, clothing, practices, landmarks, and regional identities. Results on these benchmarks show that VLMs continue to struggle with culturally grounded understanding, highlighting an important distinction: multilingual coverage does not necessarily translate into cultural understanding.

These challenges are also closely related to multimodal hallucination, where models generate plausible responses that are not sufficiently grounded in the visual input (Chen et al., 2026). In culturally situated settings, a model may rely on linguistic or cultural associations to infer an answer even when the image provides insufficient evidence. Distinguishing genuine visual understanding from such associative reasoning becomes particularly difficult when incorrect alternatives are themselves culturally plausible. Consequently, answer accuracy alone may not capture visual grounding, and evaluation should assess whether models distinguish visually supported answers from plausible but unsupported alternatives (Mousi et al., 2026b,a).

Arabic provides an important setting for examining these issues because systems must handle MSA and regional dialects while grounding predictions in culturally specific content (Al-Khalifa et al., 2025). Dallah (Alwajih et al., 2024) highlights the importance of dialect-aware Arabic multimodal modeling, while CAMEL-Bench (Ghaboura et al., 2025) shows remaining gaps in Arabic multimodal performance. However, evaluation of culturally grounded VQA and hallucination across Arabic varieties remains limited, particularly for settings where questions are presented in spoken rather than written form.

The same concerns extend to text-to-image gen-

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eration. Models may produce visually plausible images while omitting or distorting culturally important details. Standard measures of image quality and text-image alignment often fail to capture these errors or align with human judgments of cultural faithfulness (Kannen et al., 2024; Bayramli et al., 2025; Nayak et al., 2025). This is particularly relevant to Arabic and Gulf contexts, where distinct national characteristics may be reduced to generic regional representations (Elsharif et al., 2024; Almarwani et al., 2025).

To address these gaps, we introduce **IMAGEEVAL 2026**, a shared task on culturally grounded Arabic multimodal evaluation, extending the ImageEval 2025 shared-task (Bashiti et al., 2025). It brings together two complementary tasks. AYNVQA covers spoken visual question answering and hallucination detection, while CRAI-BENCH evaluates the cultural accuracy of text-to-image generation for Qatari cultural scenes. Overall, the tasks provide a unified Arabic-English evaluation of cultural grounding across multimodal understanding and generation.

**Findings:** The results highlight three main findings. ***First***, spoken VQA is more challenging in MSA than in English, with the best MSA accuracy 8.7 percentage points lower than the best English result. System analyses attribute much of this gap to Arabic ASR and the shift from synthetic training audio to human-recorded test speech. ***Second***, hallucination detection benefits from task-aware single-choice prediction. Systems that jointly select the visually grounded statement achieve substantially lower contrastive instability than independent true/-false verification, with a zero-shot system ranking first in MSA. ***Third***, CRAI-BENCH shows that competitive performance can be achieved using different combinations of visual assessment and caption-derived information. The highest-ranked submission uses a caption-version prior together with CLIP-based tie-breaking, highlighting the importance of controlling for structural cues in cultural image-evaluation benchmarks.

Overall, these findings motivate stronger multimodal grounding and cultural evaluation benchmarks that more directly measure the contribution of visual evidence.

## 2 Tasks and Datasets

The shared task considers three complementary evaluation settings: grounding spoken questions in images, distinguishing visually supported content from culturally plausible hallucinations, and assessing cultural faithfulness in generated images. This section describes the corresponding tasks, datasets, and evaluation procedures.




Figure 1: Example of Task 1: (1a) culturally grounded spoken visual QA with multiple-choice answers and (1b) image-grounded hallucination detection over true/false statements.

### 2.1 Task 1: AynVQA

AynVQA is offered as two subtasks, each in English and MSA.

- **Task 1a (Spoken VQA):** *Given an image and a spoken question with three spoken answer options, predict the index of the correct option.* Neither the question nor the options are provided as text.
- **Task 1b (Hallucination Detection):** *Given an image and three statements about it, judge each statement as true or false, where exactly one statement is grounded in the image and the other two are culturally plausible but visually unsupported.*

**Dataset.** Task 1 is derived from OASIS (Alam et al., 2025b), which pairs images with spoken and textual QA instances in English and Arabic varieties across 18 MENA countries. Figure 1 illustrates the two subtasks.

The training, development, and dev-test splits

| Split | # Examples | Source | # Ctry. | Speech |
|---|---|---|---|---|
| Train | 3,000 | OASIS | 18 | Cloned |
| Dev | 500 | OASIS | 17 | Cloned |
| Dev-test | 500 | OASIS | 17 | Cloned |
| Test | 1,000 | M$^2$CQA | 13 | Human |
| Total | 5,000 | | | |

Table 1: Task 1 data splits by source, country coverage, and speech type. All splits cover 9 cultural categories and 31 subcategories. Development-phase labels were released; dev-test and test labels were withheld until the corresponding phase closed.

are derived from OASIS, which contains multiple-choice (MCQ), open-ended (OEQ) and true/false questions. For Task 1a, we select a subset of the OASIS MCQ data for the shared task. For Task 1b, we adopt the approach proposed by Mousi et al. (2026b), converting each MCQ into three true/-false statements using GPT-4.1: one true statement corresponding to the correct answer and two false statements derived from the distractors, as illustrated in Figure 1. The test set is drawn from M$^2$CQA (Mousi et al., 2026b); each record was matched back to OASIS by image id to recover its multiple-choice question and the corresponding original human recording (Alam et al., 2025b). The public M$^2$CQA release has no overlap with the shared-task test set, and the test gold labels were released only after the test phase was over. The training, development, and dev-test splits use voice-cloned speech, whereas the test set uses human recordings, introducing variation in speakers and recording conditions. Each example is annotated with country, cultural category, and subcategory labels based on a taxonomy of nine categories and 31 subcategories, enabling fine-grained analysis across countries and cultural topics. To ensure cultural grounding and QA accuracy, a subset of the data was manually verified. Further details are provided in (Mousi et al., 2026b; Alam et al., 2025b). Table 1 summarizes the splits by size, source dataset, country coverage, and speech type.

**Evaluation.** For **Task 1a**, we evaluate systems using *accuracy*, *balanced accuracy*, and *macro-F1*. For **Task 1b**, we adopt the contrastive evaluation proposed by Mousi et al. (2026b). Each example consists of a triplet with one visually grounded statement and two plausible but unsupported statements. The primary metric, **contrastive instability (CI)**, measures whether a system makes consistent predictions across the triplet rather than evaluating each statement independently. Let $N_{\text{partial}}$ denote the number of triplets with at least one correct prediction and $N_{\text{consistent}}$ the number of these triplets for which all three statements are correctly classified. We define $CI = 1 - \frac{N_{\text{consistent}}}{N_{\text{partial}}}$. Lower CI indicates greater consistency: once a system identifies part of a triplet, it should resolve the remaining statements correctly. We additionally report *combined accuracy*, the *counterfactual hallucination rate* (CFHR), i.e., among triplets whose grounded statement is correctly accepted, the fraction in which at least one unsupported statement is also accepted, and accuracies for grounded and unsupported statements. Missing or unparseable predictions are counted as incorrect for both subtasks. Although Task 1b is a per-statement hallucination detection task, its exactly-one-true structure allowed participants to treat it as a three-way multiple-choice problem, in which case CI reduces to one minus selection accuracy and CFHR is zero by construction.

**Baseline.** We evaluate both subtasks in a zero-shot setting. For Task 1a, Qwen2.5-Omni-3B takes the image and spoken question as input and predicts the answer. For Task 1b, Qwen2.5-VL-3B takes the image and each statement as input and predicts whether the statement is grounded in the image.

### 2.2 Task 2: CRAI-Bench

CRAI-BENCH evaluates automated methods for assessing cultural accuracy in text-to-image (T2I) generation. Each instance contains (i) a **reference image** depicting an authentic Qatari cultural scene, (ii) a **caption** used to prompt a T2I model, and (iii) the corresponding **generated image**. Given these inputs, systems predict scores along five dimensions of the **Cultural Representation Accuracy Index (CRAI)**. Systems are evaluated against human-annotated CRAI scores.

**Dataset.** The CRAI-BENCH dataset consists of 40 reference images depicting authentic Qatari cultural scenes across three categories: people and traditional attire, architecture and built environment, and objects. The images were collected from open-access platforms, including Pexels, Pixabay, and Unsplash, and their cultural authenticity was verified by native Qatari cultural consultants. Figure 2 shows an example of a reference image and generated images.

Each reference image is paired with five English captions (v1 to v5) of decreasing cultural

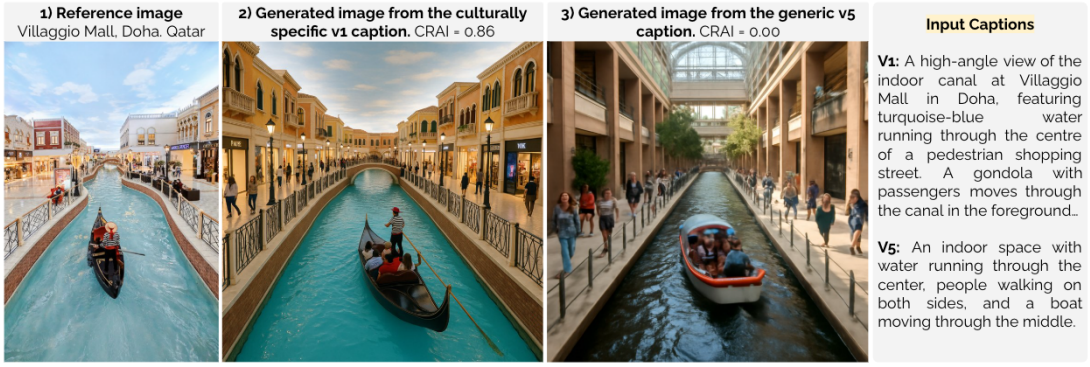



Figure 2: Example from CRAI-BENCH (Task 2). Captions with decreasing cultural specificity generate substantially different representations of the same reference scene, reflected in their human-annotated CRAI scores.

| **Dimension** | $w_d$ |
|---|---|
| **Cultural Element Accuracy** (CEA)<br>Expected cultural elements are present and correctly depicted. | $+0.30$ |
| **Contextual Coherence** (CC)<br>Elements appear in culturally appropriate settings. | $+0.20$ |
| **Cultural Specificity** (CS)<br>Specificity of the depiction to the target culture, on five levels: 1.0 uniquely Qatari, 0.75 strongly associated, 0.50 regionally common, 0.25 generic Middle Eastern, 0 none. | $+0.20$ |
| **Cultural Integrity** (CInt)<br>The representation is truthful and free of distortion. | $+0.20$ |
| **Hallucination Penalty** (HP)<br>Culturally incorrect elements not supported by the caption. Higher is worse. | $-0.10$ |
| $\mathrm{CRAI} = \sum_d w_d\, s_d = 0.30\, s_{\mathrm{CEA}} + 0.20\, s_{\mathrm{CC}} + 0.20\, s_{\mathrm{CS}} + 0.20\, s_{\mathrm{CInt}} - 0.10\, s_{\mathrm{HP}}$,<br>with $s_d \in [0, 1]$, so CRAI $\in [-0.10, 0.90]$. | |

Table 2: The five CRAI dimensions, their weights $w_d$, and the composite score. Each dimension score $s_d$ lies in $[0, 1]$; higher is better for CEA, CC, CS and CInt, and worse for HP. Cultural Integrity is abbreviated CInt to avoid collision with CI (contrastive instability).

specificity. Each caption is then used to generate one image with `gpt-image-1` at a resolution of $1024 \times 1024$ pixels, yielding 200 caption-generated image instances in total. For each instance, human annotators scored the reference image, caption, and generated image using the five CRAI dimensions, from which the composite CRAI score was computed. The same instances are also evaluated by GPT-4o as an automated baseline. The corresponding caption-version identifier ($v1$–$v5$) was included in the released instance metadata and was available to participants during evaluation. The dataset is split by reference image into 60/20/20 train, development, and test partitions, stratified by cultural category. This ensures that captions and generated images associated with the same reference image do not appear across different splits. Table 3 summarizes the split sizes.

| **Split** | **# Ref. images** | **Capt. per ref. image** | **# Inst. (gen. images)** | **Share of data** |
|---|---|---|---|---|
| Train | 24 | 5 | 120 | 60% |
| Dev | 8 | 5 | 40 | 20% |
| Test | 8 | 5 | 40 | 20% |
| Total | 40 | 5 | 200 | 100% |

Table 3: CRAI-BENCH splits. Each reference image is paired with five English captions (v1 to v5) of decreasing cultural specificity, and each caption yields one generated image, so every reference image contributes five instances. Splitting is done based on the reference image, so no reference image crosses splits.

**Evaluation.** The primary evaluation metric is Spearman correlation (Zar, 2005) ($\rho$) between predicted and human-annotated `CRAI_composite` scores, with higher values indicating stronger agreement with human rankings. Mean Absolute Error (MAE) on the composite score is used as a secondary metric and tiebreaker, with lower values being better.

The official evaluation script computes both metrics and was distributed with the starter kit. During development, participants could evaluate their systems locally using the released development labels before submitting predictions to Codabench.

**Baseline.** We use GPT-4o as the baseline system. Given the reference image, caption, generated image, and CRAI rubric, the model predicts scores for the five CRAI dimensions. The composite CRAI score is then computed using the weighted formulation defined above.

### 2.3 Competition Setup

The shared task was conducted in two phases: **development** and **test**. During the development phase, participants developed their systems using the released training and development data and sub-

| # | Team | Acc. | B-Acc. | M-F1 |
|---|---|---|---|---|
| ***English*** | | | | |
| 1 | Ahmed Ayman | 0.962 | 0.933 | 0.908 |
| 2 | NYUAD | 0.961 | 0.919 | 0.872 |
| 3 | CUET_InferX | 0.912 | 0.877 | 0.805 |
| – | *Qwen2.5-Omni-3B (baseline)* | 0.594 | 0.619 | 0.483 |
| ***MSA*** | | | | |
| 1 | NYUAD | 0.875 | 0.780 | 0.727 |
| 2 | Ahmed Ayman | 0.834 | 0.721 | 0.686 |
| 3 | Digilians | 0.656 | 0.575 | 0.504 |
| – | *Qwen2.5-Omni-3B (baseline)* | 0.191 | 0.339 | 0.169 |
| ***Reference strategies*** *(identical in both tracks)* | | | | |
| – | *Majority answer position* | 0.539 | – | – |
| – | *Random choice* | 0.333 | – | – |

Table 4: Task 1a (spoken visual QA) test results, $n = 1{,}000$ per track. Qwen2.5-Omni-3B is the official baseline. The reference strategies use no input: *majority answer position* always predicts the most frequent option, and *random choice* selects uniformly among the three options.

| # | Team | CI↓ | Acc.$_{Q+}$ | Acc.$_{Q-}$ |
|---|---|---|---|---|
| ***English*** | | | | |
| 1 | Team Falcons | 0.029 | 0.971 | 0.986 |
| 2 | Dynamos | 0.033 | 0.967 | 0.984 |
| 3 | Team Tokenizers | 0.035 | 0.965 | 0.983 |
| 4 | NYUAD | 0.041 | 0.959 | 0.980 |
| 5 | alkhder | 0.051 | 0.949 | 0.975 |
| 6 | Nile Nexus | 0.058 | 0.942 | 0.971 |
| 7 | VisionEval | 0.067 | 0.933 | 0.967 |
| 8 | keslerady* | 0.113 | 0.887 | 0.944 |
| – | *Qwen2.5-VL-3B (baseline)* | 0.267 | 0.931 | 0.863 |
| ***MSA*** | | | | |
| 1 | Ahmed Younis | 0.036 | 0.964 | 0.982 |
| 2 | NYUAD | 0.047 | 0.953 | 0.977 |
| 3 | Dynamos | 0.056 | 0.944 | 0.972 |
| 4 | alkhder | 0.096 | 0.904 | 0.952 |
| 5 | VisionEval | 0.101 | 0.899 | 0.950 |
| 6 | keslerady* | 0.142 | 0.858 | 0.929 |
| – | *Qwen2.5-VL-3B (baseline)* | 0.428 | 0.868 | 0.790 |

Table 5: Task 1b (hallucination detection) test results, ranked by contrastive instability (CI, lower is better), with accuracy on the grounded ($Q^+$) and hallucinated ($Q^-$) statements. Qwen2.5-VL-3B is the official baseline. *: no system description paper was submitted.

| # | Team | V-prior | $\rho\uparrow$ | MAE↓ |
|---|---|---|---|---|
| ***CRAI-Bench*** 5 ranked | | | | |
| 1 | VisionEval | ✓ | 0.826 | 0.147 |
| 2 | surgan_jandial* | | 0.808 | 0.151 |
| 3 | DATALabKU | | 0.804 | **0.141** |
| 4 | Ahmed Younis | ✓ | 0.781 | 0.145 |
| 5 | Ahmed Ayman | | 0.673 | 0.203 |
| – | *GPT-4o (baseline)* | | 0.519 | 0.236 |

Table 6: Task 2 (CRAI-BENCH) test results, ranked by Spearman correlation ($\rho$) between the predicted and human composite scores. The test split contains 40 instances drawn from 8 reference images, so instances are clustered within reference image, and the top four systems span only $0.045$ in $\rho$. **V-prior** marks systems that explicitly use caption-version metadata in their predictions. MAE is a secondary metric used only to break exact ties in $\rho$; no ties occurred, so it did not affect the ranking, and the lowest MAE (bold) belongs to a system ranked third. GPT-4o is the official baseline. *: no system description paper was submitted.

mitted predictions on the dev-test split to a live leaderboard. During the test phase, participants submitted predictions on the blind test set, which determined the official rankings. The test leaderboards remained hidden until the phase closed. Each team was allowed up to 16 submissions during the test phase, with a maximum of 10 submissions per day. The highest-scoring submission from each team was used to determine its official leaderboard position.

Participants could use open- or closed-source models, external data, and pretrained models, provided that all resources were disclosed in the system description paper.

All Task 1 and Task 2 data are released under CC BY-NC-SA 4.0.[1] The starter kits, evaluation scripts, and format checkers are also publicly available.[2]

## 3 Results and Discussion

In Tables 4, 5, and 6, we report the official results on the blind test sets. Overall, the results highlight three patterns: spoken VQA remains more challenging in MSA than in English, task formulation plays an important role in hallucination detection, and cultural image evaluation can be influenced by structural cues beyond the visual content.

**Task 1a (Spoken VQA).** The English track was closely matched, with the top two systems differing by only 0.001 in accuracy. MSA was more challenging, with the best accuracy reaching 0.875 compared with 0.962 for English. Analysis of results across participating systems suggests that much of this gap comes from Arabic ASR errors and differences between the synthetic training and development speech and human-recorded test speech (Appendix A). Nevertheless, all submitted systems substantially outperform the Qwen2.5-Omni-3B baseline. In MSA, the baseline also falls below the majority-position reference (0.191 vs. 0.539), high-

[1] Dataset
[2] Task resources

lighting the difficulty of the spoken Arabic setting.

**Task 1b (Hallucination Detection).** For the hallucination detection subtask, performance was strong across submitted systems, with the top three English systems differing by only 0.006 CI. All ranked systems achieved a counterfactual hallucination rate of zero, making consistency across the three statements the main differentiator. All ranked systems substantially outperformed the baseline, which evaluates each statement independently and reaches a CI of 0.267 in English and 0.428 in MSA. Notably, the top MSA system used zero-shot prompting rather than task-specific fine-tuning. These results suggest that aligning the prediction formulation with the contrastive structure of the task can be as important as model adaptation.

**Task 2 (CRAI-Bench).** For CRAI-Bench, all submitted systems outperform the GPT-4o baseline, with the top four achieving Spearman correlations between 0.781 and 0.826. The highest-ranked system, VisionEval (Sheikh, 2026), combines a caption-version prior with CLIP-based tie-breaking, while DATALabKU (AlKulaib and Jlidi, 2026) achieves the lowest MAE. Participating systems use different strategies, including caption-derived priors, vision-language judging, calibration, and score regression. Per-dimension correlations are reported in Appendix B. These results motivate closer examination of how much performance is attributable to visual evidence versus structural information provided with the task.

## 4 System Descriptions

Participating teams explored a range of approaches, including zero-shot prompting, task-specific fine-tuning, ASR-based pipelines, ensembling, and score calibration. In Table 7, we summarize the main models, adaptation strategies, and methods used across the three tasks. We briefly describe each submitted system below.

**Dynamos (Yamin et al., 2026)** The team participated in Task 1b for both English and Modern Standard Arabic (MSA). They reformulated hallucination detection as a constrained selection problem in which exactly one of three candidate statements is selected as visually grounded. Starting from zero-shot Qwen2.5-VL, they applied low-rank fine-tuning and trained multiple adapters with different random seeds. Their final system combined score averaging with an ensemble of 7B and 32B models, achieving CI scores of 0.033 for English and 0.056 for MSA.

**NYUAD (AlDahoul and Zaki, 2026)** The team addressed both spoken visual question answering and image-grounded hallucination detection. Their framework jointly processes visual and language inputs and was evaluated in both English and MSA. The study compared different language model architectures and examined the additional challenges introduced by Arabic speech, text processing, and limited culturally grounded multimodal resources.

**Nile Nexus (Zaman and Rishta, 2026)** Participating in Task 1b, the team used Qwen2.5-VL-7B-Instruct with 4-bit quantization and reformulated the task as selecting the single visually grounded statement. They also introduced confidence-based refinement through paraphrase voting. Their final system achieved a CI of 0.058 on the English track.

**DATALabKU (AlKulaib and Jlidi, 2026)** The team participated in Task 2 and proposed a reference-based framework for evaluating cultural accuracy in generated images. GPT-5.4 was used to extract cultural evidence from the reference image, caption, and generated image. These features were then mapped to CRAI scores using a lightweight regressor, improving Spearman correlation over direct GPT-5.4 scoring on the development set.

**Team Falcons (Ma et al., 2026)** The team participated in the English track of Task 1b and formulated the task as three-way image-conditioned classification. Qwen3-VL-8B was fine-tuned using 4-bit QLoRA, while uncertain predictions were checked using Gemini 3.6 Flash and GPT-5.4-mini. A prediction was changed only when both verifier models agreed, improving the final CI to 0.029 and placing the system first on the English leaderboard.

**Ahmed Ayman (Mohamed, 2026)** The team participated in Task 1a and Task 2. For spoken VQA, they examined the effects of answer representation and speech recognition, with ordinal answer labels outperforming digit-based labels. Their system achieved accuracies of 0.962 for English and 0.834 for MSA. For CRAI-Bench, they used multiple judges with calibration and tie-breaking, achieving a Spearman correlation of 0.673.

**VisionEval (Sheikh, 2026)** The team participated in Task 1b for English and MSA, as well as Task 2, using zero-shot methods. For Task 1b,

| Team | EN | MSA | Models | Adaptation | Main method |
|---|---|---|---|---|---|
| **Task 1a** Spoken Visual Question Answering 3 ranked in EN, 3 in MSA | | | | | |
| **Ahmed Ayman** | 1 | 2 | ASR GPT | ZS | ASR–LLM cascade with ordinal answer labels |
| **NYUAD** | 2 | 1 | ASR Gemini | ZS | Comparison of native-audio and ASR-based pipelines |
| **CUET InferX** | 3 | – | Qwen-Omni | FT | LoRA on the language backbone; audio and vision encoders frozen |
| **Digilians** | – | 3 | Qwen-VL ASR | ZS | ASR, constrained parsing, repair, and joint visual selection |
| **Task 1b** Image-grounded Hallucination Detection 8 ranked in EN, 6 in MSA | | | | | |
| **Team Falcons** | 1 | – | Qwen-VL Gemini GPT | FT Ens. | Fine-tuned classifier with consensus verification for uncertain cases |
| **Dynamos** | 2 | 3 | Qwen-VL | FT Ens. | Constrained selection with cross-size model ensembling |
| **Team Tokenizers** | 3 | – | Qwen-VL | FT | Answer-first prediction with structured visual checks |
| **NYUAD** | 4 | 2 | Gemini | ZS | Direct zero-shot statement verification |
| **alkhder** | 5 | 4 | Qwen-VL | ZS Ens. | Rotation-averaged selection using next-token probabilities |
| **Nile Nexus** | 6 | – | Qwen-VL | ZS Ens. | Single-choice prediction with confidence-based paraphrase voting |
| **VisionEval** | 7 | 5 | Qwen-VL | ZS | Forced-choice prediction from candidate-token scores |
| **keslerady*** | 8 | 6 | – | – | *Not reported* |
| **Ahmed Younis** | – | 1 | Gemini | ZS | Selection of the single visually grounded statement |
| **Task 2** CRAI-Bench single track, 5 ranked | | | | | |
| **VisionEval** | 1 | | CLIP | Ens. | Caption-version prior with CLIP-based tie-breaking |
| **surgan_jandial*** | 2 | | GPT | ZS Ens. | VLM judging with calibration and re-evaluation of near ties |
| **DATALabKU** | 3 | | GPT | ZS Ens. | Cultural evidence extraction followed by score regression |
| **Ahmed Younis** | 4 | | Gemini | ZS Ens. | Caption-version prior combined with vision-based judges |
| **Ahmed Ayman** | 5 | | GPT Claude | ZS Ens. | Multiple judges with score calibration and tie-breaking |

Table 7: Overview of submitted ImageEval 2026 systems. Figures in the EN and MSA columns give the team's position in the official ranking, badged for the top three; the field size for each track is stated in the task band, so a position should be read against it. A dash marks a track the team did not enter. **Adaptation** is ZS (zero-shot prompting), FT (task-specific fine-tuning) and Ens. (ensembling, voting, or score calibration). Every system ranked in Tables 4 to 6 appears here. Across all entries, models used included Qwen-VL 7, Gemini 5, GPT 5, ASR 3, and Qwen-Omni, Claude and CLIP 1 each. *: no system description paper was submitted; where available, details were taken from the submission form.

they replaced independent true/false predictions with a single three-way forced-choice formulation based on candidate-token logits. For CRAI-Bench, they used a caption-version prior with tie-breaking based on frozen CLIP features. The resulting system achieved a Spearman correlation of 0.8258 and ranked first on Task 2.

**Team Tokenizers (Hoque and Hossain, 2026)** Participating in the English track of Task 1b, the team proposed HALDETECT, which treats the three candidate statements as a single contrastive grounding decision. Their best system fine-tuned Qwen2.5-VL-7B-Instruct using 4-bit QLoRA while keeping the vision encoder frozen. The system achieved a CI of 0.035 and ranked third on the English leaderboard.

**alkhder (Pasa et al., 2026)** The team participated in Task 1b for both English and MSA using a pre-trained 7B vision–language model in a zero-shot setting. They selected the single visually supported statement using next-token log-probabilities and reduced positional bias through a Latin-square rotation over candidate orderings. The system achieved CI scores of 0.051 for English and 0.096 for MSA.

**CUET InferX (Semon and Islam, 2026)** The team participated in the English track of Task 1a. They fine-tuned Qwen2.5-Omni-3B using LoRA on the language reasoning backbone while keeping the audio and vision encoders frozen. The resulting end-to-end audio–visual system achieved 0.912 test accuracy and ranked third in the English track.

**Ahmed Younis (Younis, 2026)** The team participated in the MSA track of Task 1b and in Task 2. For CRAI-Bench, they combined a frequency-based prior with two vision-based judges and optimized the blending weights against Spearman correlation, achieving 0.7806 on the test set. For Task 1b, they reformulated the task as selecting the single grounded statement, achieving a CI of 0.036 and ranking first on the MSA track.

**Digilians (Hassan et al., 2026)** The team participated in the MSA track of Task 1a using a modular spoken VQA pipeline. Their approach first performed speech recognition, then extracted the question and answer choices before applying visual multiple-choice reasoning. A larger speech recognition model was used only when the initial transcript was unreliable. The system achieved an accuracy of 0.656 on the blind test set.

## 5 Related Work

**Spoken Visual Question Answering.** Prior work on spoken QA has examined the effects of ASR errors, conversational context, and multilingual speech (Lee et al., 2018; You et al., 2022; Alam et al., 2025a; Ali et al., 2026; WANG et al., 2026; Sara et al., 2026). These benchmarks, however, do not require visual grounding. More recent datasets such as TM-PathVQA and TM-VQA combine spoken questions with images across multiple languages (Rajkhowa et al., 2024; Chowdhury et al., 2025), but rely mainly on synthesized speech and do not target culturally grounded interaction.

Arabic multimodal QA has so far focused largely on image-text settings. VAQA, CAMEL-Bench, PEARL, JEEM, and AraVQA cover domains, cultural knowledge, dialects, and factoid questions (Kamel et al., 2023; Ghaboura et al., 2025; Alwajih et al., 2025; Kadaoui et al., 2026; Alrowili et al., 2026). OASIS extends this line of work to speech with 3.7M spoken questions across 18 Arab countries, covering MSA and dialectal Arabic (Alam et al., 2025b).

**Hallucination Benchmarks and Metrics.** Vision-language hallucination is commonly divided into *faithfulness hallucination*, where outputs conflict with the visual input, and *factuality hallucination*, where they contradict external knowledge (Chen et al., 2026). ImageEval focuses on faithfulness hallucination. Existing benchmarks such as POPE, ROPE, FGHE, RAH-Bench, R-Bench, and AutoHallusion evaluate unsupported objects, attributes, relations, and spatial setups (Chen et al., 2024; Wang et al., 2024; Wu et al., 2024). Broader suites include LongHalQA, AMBER, and MERLIM (Qiu et al., 2024; Wang et al., 2023; Villa et al., 2025), while CHAIR, OpenCHAIR, CC-EVAL, and NOPE evaluate hallucination in open-ended generation (Rohrbach et al., 2018; Ben-Kish et al., 2024; Zhai et al., 2023; Lovenia et al., 2024).

Most of these benchmarks use accuracy, F1, or object-level precision and recall. Such measures do not clearly distinguish visual-recognition failures from cases where a model recognizes the image correctly but accepts a plausible unsupported alternative. Many also rely on English-language or Western-centric resources such as MS COCO (Lin et al., 2014). Although CVQA introduces culturally diverse visual questions (Romero et al., 2024), hallucination evaluation for Arabic and the MENA region remains limited (Alansari and Luqman, 2025; Mousi et al., 2026b).

**Cultural Evaluation of Text-to-Image Generation.** Recent text-to-image benchmarks increasingly evaluate cultural accuracy in addition to visual quality and prompt alignment. CUBE, CULTDIFF, CulturalFrames, and RusCode examine culturally salient content across countries and settings (Kannen et al., 2024; Bayramli et al., 2025; Nayak et al., 2025; Vasilev et al., 2025). CULTIVate and MOSAIG extend this direction to social activities and multicultural scenes (Malakouti et al., 2026; Bhalerao et al., 2026). Related efforts include the Cultural Relevance Index for Arabic culture (Elsharif et al., 2024), its use in Ara-Pic (Elsharif et al., 2025), CAIRe's knowledge-grounded cultural judgments (Yayavaram et al., 2026), and human-centered evaluation rubrics (Johnson et al., 2026). ImageEval 2025 further advances culturally grounded multimodal evaluation for Arabic through a shared task on image captioning, introducing an open-source, manually captioned dataset and evaluating Arabic image captioning systems using automatic metrics, LLM-based judgments, and human assessment (Bashiti et al., 2025).
**ImageEval 2026** builds on these efforts by bringing culturally grounded understanding and culturally faithful generation into a unified Arabic- and MENA-centered evaluation setting. It combines bilingual English–MSA evaluation for spoken visual QA and image-grounded hallucination detection with evaluation of cultural fidelity in generated images depicting culturally-specific scenes.

## 6 Conclusion

We presented ImageEval 2026, a shared task on culturally grounded Arabic multimodal evaluation. The shared task combines AynVQA, which evaluates spoken visual question answering and image-grounded hallucination detection in English and MSA, with CRAI-Bench, which evaluates cultural accuracy in generated images. A total of 14 teams participated in the test phase, with 12 teams submitting system description papers. The results highlight three main findings. First, spoken VQA remains more challenging in MSA than in English, with Arabic ASR contributing substantially to the performance gap. Second, hallucination detection benefits from task-aware single-choice formulations that directly identify the vi-

sually grounded statement. Third, CRAI-BENCH shows that caption-specificity cues can strongly influence cultural evaluation, allowing systems with limited use of visual evidence to perform well. These findings motivate further work on robust Arabic speech processing, visually grounded hallucination detection, and cultural evaluation methods that rely more directly on image content. We release the datasets and evaluation resources to support future research on culturally grounded Arabic multimodal systems. For future work, we plan to expand the benchmark with broader Arabic dialect coverage, more diverse speech and image-generation conditions, and stronger controls against structural cues.

## Limitations

For Task 1a, the training, development, and dev-test audio is synthesized, while the test set uses human recordings, introducing a challenge in speech conditions that particularly affects MSA performance. Task 1b uses a constrained setting with exactly one grounded statement and two unsupported alternatives, so the results may not generalize to less structured hallucination scenarios. Task 2 is limited to Qatari cultural content and contains 40 reference images and 200 generated images, all produced using a single text-to-image model. Its five caption versions also introduce a strong association between caption specificity and human scores, which allowed systems using caption-version information to perform well with limited use of visual evidence. Future editions should broaden the cultural and geographic coverage, include more varied speech and generation sources, and reduce structural cues that can be exploited without fully evaluating the image; for Task 1b in particular, varying the number of grounded statements per example (zero, one, or two) would require systems to judge each statement independently rather than selecting a single answer.

## Broader Impact and Ethical Considerations

ImageEval 2026 aims to improve the evaluation of multimodal systems for Arab cultural contexts across speech, text, and image generation. By releasing shared data, evaluation scripts, and baselines, the benchmark can lower the barrier to developing and comparing Arabic multimodal systems and help identify weaknesses that may otherwise remain hidden in predominantly English-centric evaluation. Our results highlight notable gaps, particularly in Arabic speech recognition and the cultural accuracy of generated images, which may affect the reliability and inclusiveness of systems deployed for Arabic-speaking users. At the same time, benchmark scores should not be interpreted as a complete measure of cultural competence, as Arab societies are linguistically, geographically, and culturally diverse. We therefore encourage future work to expand coverage across countries, dialects, communities, and cultural perspectives, and to use the benchmark as one component of broader human-centered evaluation. We do not identify any significant ethical concerns associated with the benchmark beyond these limitations. The datasets and resources are released under their respective licenses and are intended for research purposes.

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## A Development-Phase Results

Tables 10, 9, and 8 report the development-phase results (each team's latest submission). For Task 1, we evaluated submissions on the dev-test split, using synthetic TTS audio for Task 1a; for Task 2, we used the development split. The results show several patterns. For Task 1a, English systems generally outperform their MSA counterparts, with a wider performance spread in MSA. Task 1b shows strong performance among the leading systems in both languages, although MSA remains more challenging overall. For Task 2, both participating systems substantially outperform the GPT-4o baseline. Overall, the development results indicate stronger performance in English for spoken VQA and consistent improvements over the released baselines across all tasks.

| # | Team | VP | $\rho \uparrow$ | MAE$\downarrow$ |
|---|---|---|---|---|
| ***CRAI-Bench*** 2 ranked | | | | |
| 1 | Ahmed Younis | ✓ | 0.708 | 0.188 |
| 2 | VisionEval | ✓ | 0.641 | 0.204 |
| – | *GPT-4o (baseline)* | | 0.215 | 0.367 |

Table 8: Task 2 development-phase results on the dev split, ranked by Spearman correlation ($\rho$) of the composite score. The dev split contains 40 instances drawn from 8 reference images. **VP** marks systems that use a caption-version prior; both submitted systems do. Only two teams submitted to this leaderboard, and the development ranking did not carry over to the test set, where VisionEval placed first and Ahmed Younis fourth (Table 6). GPT-4o is the official baseline.

## B CRAI-Bench Per-Dimension Results

Table 11 reports the test-phase Spearman correlations for each CRAI dimension. The results show that the hallucination-penalty (HP) dimension is consistently more difficult than the other dimensions across submitted systems. HP correlations range from $-0.046$ to $0.301$, while correlations for Cultural Element Accuracy (CEA), Contextual Coherence (CC), Cultural Specificity (CS), and Cultural Integrity (CInt) are generally substantially higher.

Across the five submitted systems, HP is the weakest-correlated dimension in every case. The remaining CRAI dimensions show considerably stronger agreement with the human annotations, although their relative difficulty varies by system. For example, VisionEval achieves its strongest dimensional correlations on CC and CInt, whereas

| # | Team | CI↓ | Acc.$_{Q+}$ | Acc.$_{Q-}$ |
|---|---|---|---|---|
| ***English*** 7 ranked | | | | |
| 1 | Team Tokenizers‡ | 0.028 | 0.972 | 0.986 |
| 2 | NYUAD‡ | 0.028 | 0.972 | 0.986 |
| 3 | Team Falcons | 0.030 | 0.970 | 0.985 |
| 4 | nawwad | 0.036 | 0.964 | 0.982 |
| 5 | Nile Nexus | 0.050 | 0.950 | 0.975 |
| 6 | ameya | 0.284 | 0.716 | 0.858 |
| 7 | VisionEval† | 0.313 | 0.940 | 0.839 |
| – | *Qwen2.5-VL-3B (baseline)* | 0.313 | 0.940 | 0.839 |
| ***MSA*** 4 ranked | | | | |
| 1 | NYUAD | 0.036 | 0.964 | 0.982 |
| 2 | NAMAA Community | 0.046 | 0.954 | 0.977 |
| 3 | lettycat | 0.082 | 0.918 | 0.959 |
| 4 | VisionEval† | 0.490 | 0.834 | 0.773 |
| – | *Qwen2.5-VL-3B (baseline)* | 0.490 | 0.834 | 0.773 |
| ***Reference*** identical in both tracks | | | | |
| – | *Random selection* | 0.667 | 0.333 | 0.667 |

Table 9: Task 1b development-phase results on the dev-test split ($n = 500$ triplets per track), ranked by contrastive instability (CI, lower is better), with accuracy on the grounded ($Q^+$) and unsupported ($Q^-$) statements. Qwen2.5-VL-3B is the official baseline; *random selection* picks one of three statements uniformly. All ranked systems except VisionEval use the single-choice formulation of Table 5, under which all three metrics follow from CI. †Identical to the baseline output. ‡Exact tie; ranked by submission time.

| # | Team | Acc. | B-Acc. | M-F1 |
|---|---|---|---|---|
| ***English*** 3 ranked | | | | |
| 1 | NYUAD | 0.974 | 0.974 | 0.974 |
| 2 | romey101 | 0.972 | 0.972 | 0.972 |
| 3 | CUET_InferX | 0.916 | 0.916 | 0.916 |
| – | *Qwen2.5-Omni-3B (baseline)* | 0.664 | 0.666 | 0.661 |
| ***MSA*** 3 ranked | | | | |
| 1 | NYUAD | 0.920 | 0.919 | 0.921 |
| 2 | Digilians | 0.784 | 0.780 | 0.782 |
| 3 | VisionEval† | 0.398 | 0.410 | 0.344 |
| – | *Qwen2.5-Omni-3B (baseline)* | 0.398 | 0.410 | 0.344 |
| ***Reference*** identical in both tracks | | | | |
| – | *Random choice* | 0.333 | 0.333 | – |

Table 10: Task 1a development-phase results on the dev-test split ($n = 500$ per track, synthetic TTS audio). Qwen2.5-Omni-3B is the official baseline; *random choice* selects uniformly among the three options. The three metrics coincide for the ranked English systems because the dev-test answer positions are balanced, so at high accuracy per-class recall and precision converge; the test split instead has a majority position of 0.539 (Table 4), and this difference in label balance is one of the conditions that changes between the two phases. †Submission identical to the baseline output on all three metrics.

| Team | CEA | CC | CS | CInt | HP |
|---|---|---|---|---|---|
| VisionEval | 0.783 | 0.815 | 0.631 | 0.805 | 0.111 |
| surgan_jandial | 0.755 | 0.782 | 0.695 | 0.762 | 0.184 |
| DATALabKU | 0.766 | 0.716 | 0.724 | 0.733 | 0.301 |
| Ahmed Younis | 0.750 | 0.758 | 0.692 | 0.708 | 0.000 |
| Ahmed Ayman | 0.764 | 0.489 | 0.469 | 0.391 | -0.046 |

Table 11: Task 2 test-phase Spearman correlations by CRAI dimension. Higher correlation indicates stronger agreement with human annotations. For HP, higher underlying scores indicate worse cultural hallucination, while higher Spearman correlation still indicates better agreement with human annotations.

DATALabKU shows comparatively balanced performance across CEA, CC, CS, and CInt. These results support the observation that predicting hallucination penalties remains particularly challenging for current CRAI-Bench systems.